Calculations report: TEST-POOL-01 (assessment test, not a real project)

Job number: TEST-0001

Customer: TEST customer

NOT APPROVED: issued for review only.

The design rules are this software's defaults; the company has not confirmed them.

Basis of calculation

- Design current Ib = P / (k U cos phi); breaker In >= Ib, cable Iz >= In (IEC 60364-4-43 §433.1; ABB Electrical Installation Handbook Vol. 2, §2.1-2.3).
- Current-carrying capacity from IEC 60364-5-52 Annex B, corrected for ambient temperature and grouping.
- Voltage drop from the origin, from Schneider EIG Fig. G28 (copper) and ABB §2.2.2 (aluminium), against IEC 60364-5-52 Annex G or the company's limits.
- Fault level at a sub-board through its feeder, IEC 60909-0 (cmax 1.10); breaking capacity at or above it.
- Earth fault disconnection, IEC 60364-4-41 §411.4.4: Zs x Ia <= 0.95 U0, Ia the top of the breaker's instantaneous band (IEC 60898-1).
- Short circuit along the cable, ABB §2.4: the far-end Ikmin trips the breaker at once; k^2S^2 from Table 1 for comparison with the breaker's let-through I^2t.

Board TEST-DB-POOL

400 V, 3-phase, TN-S · Ik 10 kA · Ze 0.35 ohm · Incomer C63 4P 10 kA

Circuit	Ib A	Protection	Cable	L m	dU %	Zs ohm	Zs max	k^2S^2 $(kA)^2s$
Underwater light	0.6	C10 1P 10 kA	3G1.5 Cu PVC	-	-	-	2.194	0.03
Plant room socket	1.4	C16 1P 10 kA	3G2.5 Cu PVC	-	-	-	1.372	0.083
Water heater	14.4	C16 1P 10 kA	3G1.5 Cu PVC	-	-	-	1.372	0.03
Filtration pump	12	D13 3P 10 kA	5G1.5 Cu PVC	-	-	-	0.844	0.03

Notes

- Group 1 breaker raised to 32 A, 1.6 x the 16 A breaker after it, for discrimination.
- Phase imbalance 52.3 % exceeds the company's 10 %: the loads cannot be spread more evenly as given.
- Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination.
- Breakers are rated for at least 10 kA breaking capacity against 10 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too.
- Leave 1 spare outgoing ways (20 %).
- Loads without a power factor were taken at cos phi 0.9.

- Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.

- 4 circuit(s) have no cable length given; their voltage drop is not checked.

- 1 circuit(s) are not checked for earth fault disconnection: no cable length, a motor starter's breaker, or a breaker not selected here.

- Each cable's withstand k^2S^2 is in the cable list; compare it with the selected breaker's let-through I^2t at 10 kA from its maker's curve (ABB handbook §2.4).

- Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.

- Designed for Palestine, under IEC 60364 (adopted by the Palestine Standards Institution): its stated limits apply beneath the company's own settings.

- Circuit rules are this software's defaults, not confirmed by the company's engineers.

- Enclosure: ABB Mini Center compact GDMS348 RX (2CVA360001P0001), 3 rows of 16 modules, H x W x D 648 x 400 x 117 mm, IP41, busbar 200/250 A, 35 kA; 31 modules are left for spare ways (ABB Distribution Boards catalogue (1SKC802027C0201), Mini Center compact, pp. 5, 7). The outgoing terminal strip is not counted.

Prepared	Checked	Approved